THE YAMAGUCHI FRAMEWORK

What Preserves Zero

Collected Papers, Volumes I–XV
Reach and Limits of a Spectral Analyzer across the Seven Millennium Problems

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This collection claims a proof of none of the open Millennium Prize Problems.
What each volume claims is stated under the markers VERIFIED CLASSICAL CONJECTURE UNVERIFIED REFUTED.

Preface — Editorial Policy and Markers

This volume collects the Yamaguchi Framework, Volumes I through XV, each condensed to a short paper that preserves the main theorems, the skeleton of the proofs, the numerical verification, and the verification status of the original. Volumes I–V are not reproduced in their original form; for them we record only the present status of their claims in the light of the later volumes (retrospective notes). Volumes VI–XV transcribe the main results faithfully from the originals. Volume XV, the capstone, is included here in full.

The editorial policy is a single rule: **separate, inside one and the same computational system, what has been proved from what has not**, using the markers VERIFIED (proof and numerical verification given in the original volume), CLASSICAL (citation of an established result due to others), CONJECTURE (precise but unproved), UNVERIFIED (lacking a definition or a verification), and REFUTED (established as false, or failing as a proof). No paper in this collection claims a proof of Navier–Stokes regularity, of the Riemann Hypothesis, of P ≠ NP, of the Hodge Conjecture, of Yang–Mills existence and mass gap, or of the Birch–Swinnerton-Dyer Conjecture. What each paper claims is where a spectral analyzer — a device that reads poles, finite parts, and zero modes off a sequence of eigenvalues — reaches on each problem, and where it stops.

The technical skeleton common to all volumes is threefold. (1) For a closed manifold, or for any non-negative operator with discrete spectrum, the spectral zeta function $\zeta(s) = \Sigma_{\lambda>0}\, \lambda^{-s}$ and its Mellin representation $\Gamma(s)\zeta(s) = \int_0^\infty t^{s-1}(\mathrm{Tr}\ e^{-t\Delta} - n_0)\ dt$. (2) Its readouts: the rightmost pole $s = d/2$ (Weyl law); the value at the origin $\zeta(0) = a_1 - n_0$ (topology and zero modes); the slope at the origin $\zeta'(0)$ (geometry, log-determinant); the odd part of the functional equation (deviation from Poisson symmetry); the behavior at infinity $\lim \zeta(s)^{-1/s} = \lambda_1$ (spectral gap); and — added in Volume XV — the order of the zero at the center of the critical line. (3) What cannot be read: the spectrum does not determine the metric (Milnor), the nonlinear term (Navier–Stokes), the Euler product (Riemann), the clause structure (SAT), or the complex structure (Hodge); in four-dimensional Yang–Mills the operator itself has not yet been constructed; and a numerical zero is never a proof that a value is zero (BSD).

All numerical verification is by Python 3 (standard library, numpy, or mpmath), IEEE double precision unless stated otherwise, executed in a Termux-compatible environment; reproduction code is attached to each original volume, and the BSD verification code of Volume XV is published alongside this collection (`vol15_bsd_verification.py`).

I	Symbol-Shape Entropy (retrospective note)
II	The Corrected Beta–Zeta Equation and Language Generation (retrospective note)
III	Applications to Quantum Computation (retrospective note)
IV	Zeta, Entropy, Generation (retrospective note)
V	Manifold Integration and Quantum Engineering (retrospective note)
VI	The Structural Correspondence between Beta and Zeta
VII	The Quotient-Algebraic Logarithmic Structure of Beta and Euler's Constant
VIII	Quantizing Gauss's Theory of Surfaces
VIII-S	Supplement: From Gauss to Thurston–Perelman (Ricci Flow and the Eight Geometries)
IX	The Partial-Integration Analyzer of the Gamma Function and the Gravity/Antigravity Branches
X	The Rightmost Pole of the Spectral Zeta and Navier–Stokes Regularity
XI	The Riemann Zeta as the Spectral Zeta of the Circle
XII	Satisfiability as the Number of Zero Modes and P ≠ NP
XIII	Zero Modes of Harmonic Forms and the Hodge Conjecture
XIV	The Mass Gap as the s → ∞ End of Zeta and the Yang–Mills Problem
XV	The Birch–Swinnerton-Dyer Conjecture as the Zero at the Center of the Critical Line, and a Synthesis of the Seven Millennium Problems (in full)
—	Synthesis: the Readout Table and the Six Walls / Appendix: Index of Propositions / References

VOLUME I · RETROSPECTIVE NOTE

Symbol-Shape Entropy

Abstract. The point of departure of the framework: the idea of assigning a Shannon-type entropy x log x to the shapes of symbols. This note does not reconstruct the original formulation; it records the status of its claims in the light of the later volumes.

Claim	Status	Verdict in later volumes
x log x is the integrand kernel of the entropy $-\Sigma p_i \log p_i$ of a probability distribution	CLASSICAL	Shannon 1948
An entropy of symbol shapes unifies prime distribution, curvature, and language generation	UNVERIFIED	The unifying equation $\zeta = \beta/\log x$ is false (Vol. VI)
x log x has an exact meaning as the solution entropy of SAT, $s(\alpha) = \log_2(\#SAT)/n$	VERIFIED	Vol. XII, Theorem 3
The s-derivative kernel $t^{s-1}\log t$ of the Mellin representation reads $\zeta'(0)$ (geometry)	VERIFIED	Vol. IX, Theorem B

What survives is the classical fact that x log x is an entropy kernel, and that it later acquired two rigorous roles (the SAT solution entropy; the logarithmic kernel of Γ'). No new mathematical claim original to Volume I survives in verified form.

VOLUME II · RETROSPECTIVE NOTE

The Corrected Beta–Zeta Equation and Language Generation

Abstract. The program of using a "corrected" form of $\zeta(s) = \beta(p,q)/\log x$ as the weight function of a language generator. The central identity was established as false, as a pointwise identity, in Volume VI.

Claim	Status	Verdict
$\zeta(s) = \beta(p,q)/\log x$ (pointwise identity)	REFUTED	Fails at every tested point, including prime arguments (mpmath, 30 digits; Vol. VI §5)
Read as a defining equation: unique solvability $x(s) = e^{W(\zeta(s))}$ (Lambert W), threshold $s^* = 1.3942\ldots$	VERIFIED	Vol. VI §6; impossibility theorem at prime arguments
The language generator's output depends on the equation	UNVERIFIED	The generator depends on n-gram-type counting, not on the equation

There is no theorem to record from this volume. What is to be recorded is the definitive refutation of the central identity (Vol. VI, Theorem 0), and the fact that this refutation became the premise of every later paper in the collection.

VOLUME III · RETROSPECTIVE NOTE

Applications to Quantum Computation

Abstract. An implementation of Grover-type amplitude amplification. The implementation is correct, but by known lower bounds it does not bring NP problems into polynomial time (Vol. XII, Theorem 4).

> VERIFIED CLASSICAL **Theorem III.1** (Grover 1996; BBBV 1997). Among $N = 2^n$ candidates with n_0 solutions, Grover iteration finds a solution with probability ≈ 1 in $k \approx (\pi/4)\sqrt{(N/n_0)}$ oracle calls; any quantum algorithm that identifies solutions only through the oracle requires $\Omega(\sqrt{(N/n_0)})$ calls. Measured (Vol. XII, Table 3): $n = 8, 10, 12$ give theoretical $k^* = 13, 9, 21$ against measured best $k = 12, 9, 20$, success probability ≥ 0.9999.

The exponent 2^n becomes $2^{n/2}$; it does not become polynomial. This is the range within which the circuits of Volume III may claim to "solve" anything.

VOLUME IV · RETROSPECTIVE NOTE

Zeta, Entropy, Generation

Abstract. The program of unifying the three concepts. No unifying equation exists; each of the three later found its own rigorous home.

Concept	Rigorous home in later volumes	Status
Zeta	The spectral zeta of the circle S^1: $\zeta_{S^1}(s) = 2\zeta_R(2s)$ (Vol. XI, Thm. 1)	VERIFIED
Entropy	The SAT solution entropy $s(\alpha)$ (Vol. XII, Thm. 3)	VERIFIED
Generation	No corresponding rigorous formulation	UNVERIFIED

VOLUME V · RETROSPECTIVE NOTE

Manifold Integration and Quantum Engineering

Abstract. The program of connecting manifold integration to quantum engineering through the notations "global differential manifold" and "global partial integration" (Y1)(Y2). The parts that admit a meaning reduce to the partial integration of the Gamma function.

Notation	Reduction	Status
Global quotient algebra (Y1)	A "verifiable shadow" of the logarithmic derivative rule $(\log F/G)' = F'/F - G'/G$ (Vol. VII, Conj. 1)	CONJECTURE
Global partial integration (Y2)	$\Gamma(s+1) = [-e^{-x}x^s]_0^\infty + s\Gamma(s)$; the boundary term t^s/s is the pole $1/s$ of the Mellin representation (Vol. IX)	VERIFIED (after reduction)
$H\Psi = \oplus(i\hbar\nabla)\oplus L$	No domain, no action, no self-adjointness given	UNVERIFIED
$(d/df)F = e^f + e^{-f} \geq e^f - e^{-f}$ (gravity/antigravity)	The cosh/sinh decomposition of the Mellin transform; each branch diverges separately (Vol. IX, Prop. A)	REFUTED (as a definition of branches)

The Structural Correspondence between Beta and Zeta — Functional Equation, Trivial Zeros, Critical Line

Abstract. The pointwise identity $\zeta(s) = \beta(p,q)/\log x$ is false. But the Beta function does appear, exactly, inside Riemann's functional equation, and from there the trivial zeros and the axis property of the critical line follow. All numerics: mpmath, 30 digits.

REFUTED **Theorem VI.0.** $\zeta(s) = \beta(p,q)/\log x$ is false as a pointwise identity: it fails at every tested point, including prime arguments. Read as a defining equation it has the unique solution $x(s) = e^{W(\zeta(s))}$, with solvability threshold $s^* = 1.3942\ldots$, and is impossible at prime arguments.

VERIFIED **Theorem VI.1** (Beta-type functional equation). For $s \notin \mathbb{Z}$, $s \neq 1$,

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \cdot \beta(s, 1-s)/\Gamma(s) \cdot \zeta(1-s).$$

Proof. Substitute $\beta(s,1-s) = \Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$ into Riemann's functional equation. ∎ Verification: at $s = 2.5, -1.5, 3+2i, 2+7i, \tfrac{1}{2}+14.1347i, \tfrac{1}{2}+21.0220i$ the two sides differ by $\leq 6.5\times10^{-31}$.

VERIFIED **Corollary VI.2** (trivial zeros). $\zeta(-2n) = 0$ for $n \geq 1$, from the simple zeros of $\sin(\pi s/2)$ and the finiteness of $\Gamma(1+2n) = (2n)!$. **Corollary VI.3** (the critical line is an axis). $\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfies $\xi(s) = \xi(1-s)$; the fixed line is $\mathrm{Re}\, s = \tfrac{1}{2}$. That the zeros lie on this line (RH) does not follow from this theorem.

The Quotient-Algebraic Logarithmic Structure of Beta and the Appearance of Euler's Constant through Zeta Differences

Abstract. In the expansion of log B = log Γ + log Γ − log Γ, Euler's constant γ cancels exactly at first order and is recovered along the routes of zeta-difference series and digamma differences. mpmath, 50 digits. A missing term (− log(1+α+β)) in the output of a conversational AI is corrected.

VERIFIED CLASSICAL **Theorem VII.1.** For $|\alpha+\beta| < 1$,

$$\log B(1+\alpha, 1+\beta) = -\log(1+\alpha+\beta) + \Sigma_{n\geq 2} (-1)^n \zeta(n)/n \cdot [\alpha^n + \beta^n - (\alpha+\beta)^n],$$

with the first-order γ-terms cancelling exactly: $-\gamma\alpha - \gamma\beta + \gamma(\alpha+\beta) = 0$. Verification: $\alpha = 0.2$, $\beta = 0.3$ gives residual 6.7×10^{-52}; the version with the missing term has residual exactly log 1.5 = 0.4054651081….

VERIFIED CLASSICAL **Theorem VII.2.** $\gamma = \Sigma_{n\geq 2}(-1)^n\zeta(n)/n$ (exact to 50 digits); $\psi(p)+\psi(q) - \psi(p+q)|_{p=q=1} = -\gamma - 1$; $2\psi(½) - \psi(1) = -\gamma - 4\log 2$.

CONJECTURE **Conjecture VII.3** (global partial-integration manifold). A manifold structure pairing the kernel log x of Γ′(s) with the boundary term of Γ(s+1) = sΓ(s) under (Y2). In Volume IX the boundary term was made rigorous as the pole 1/s of the Mellin representation (a topological invariant).

Quantizing Gauss's Theory of Surfaces — the Euler Characteristic Is the Value of the Spectral Zeta at s = 0, and the Shape Is Not Determined

Abstract. For a closed connected surface, $\chi(M) = 6(\zeta_\Delta(0)+1)$. Topology ($\zeta(0)$) and geometry ($\zeta'(0)$) separate inside the spectrum. By Milnor's 16-dimensional isospectral pair, the map space ↦ spectrum is not injective. Gravity has not been quantized.

VERIFIED **Theorem VIII.1** (heat-kernel Gauss–Bonnet). Tr $e^{-t\Delta} \sim \text{Area}/(4\pi t) + a_1 + O(t)$, with $a_1 = (1/12\pi)\int K\, dA = \chi/6$. Measured on S^2: 0.3333307744 (error 7.7×10^{-6}).

VERIFIED **Theorem VIII.2** (main theorem). $\chi(M) = 6(\zeta_\Delta(0) + 1)$. Proof. The s = 0 residue reading of the Mellin representation gives $\zeta(0) = a_1 - n_0$, and $n_0 = 1$ for connected M. ∎ Verification: S^2 ($\chi = 2$, deviation 6.2×10^{-6}); five flat tori ($\chi = 0$, deviation $\le 2.7\times10^{-5}$, independent of aspect ratio).

VERIFIED **Theorem VIII.3** (separation). Under $g \to a^2 g$, $\zeta_a(s) = a^{2s}\zeta_1(s)$: the value $\zeta(0)$ is invariant (topology), the slope $\zeta'_a(0) = 2 \log a \cdot \zeta_1(0) + \zeta'_1(0)$ moves (geometry). Measured: radii 0.5–5 keep $\chi = 2$ while $\det_\zeta$ runs 1.268 → 27.32; tori of fixed area and aspect ratio 1 → 5 keep $\chi = 0$ while log det runs 2.621 → 0.0495.

VERIFIED CLASSICAL **Theorem VIII.4** (Milnor 1964). $R^{16}/(E_8 \oplus E_8)$ and R^{16}/D_{16}^+ are isospectral and non-isometric; the theta-series coefficients agree exactly for norms 0–16 (1, 480, 61920, 1050240, 7926240, …). The unit of quantization is not the manifold but the isospectral class.

Walls. (a) background independence, (b) the constraint algebra, (c) non-renormalizability — unmoved; (d) isospectrality — added by this volume.

From Gauss to Thurston–Perelman — Ricci Flow, the Eight Geometries, and the Exact Location of the Homomorphism

Abstract. The two-dimensional result of Volume VIII collapses in three dimensions, and the collapse points to the true contact between Ricci flow and spectra. This supplement is the volume of the collection that touches the one solved Millennium problem, the Poincaré Conjecture.

VERIFIED **Theorem S.1.** For every closed connected 3-manifold, independently of the metric, $\zeta_\Delta(0) = -1$. Proof. The odd Seeley–DeWitt coefficients vanish on closed manifolds; $d = 3$ is odd, so $a_3 = 0$, and $n_0 = 1$. ∎ Verified on three different topologies — S^3, flat T^3, $S^2\times S^1$ — all giving -1 (errors $2.2–3.7\times10^{-6}$). In three dimensions $\zeta(0)$ carries no topological information: this is a spectral restatement of why Thurston's geometrization is needed.

VERIFIED **Theorem S.3** (Perelman's λ is an eigenvalue). With $u = e^{-f/2}$, Perelman's functional $F(g,f) = \int(R + |\nabla f|^2)e^{-f}dV$ becomes $F = \int(4|\nabla u|^2 + Ru^2)dV$, so $\lambda(g)$ = the least eigenvalue of the operator $-4\Delta + R$; on homogeneous geometries $\lambda(g) = R$. **The monotone quantity of Ricci flow is an eigenvalue.**

VERIFIED **Theorems S.4–S.5.** Of Thurston's eight model geometries, exactly three are Einstein — E^3, S^3, H^3 (exact rational computation via Milnor's frame formulas) — and these are exactly the fixed points of the normalized Ricci flow. Numerical integration of the flow: $R = \lambda$ is monotone non-decreasing in every case tested (Perelman monotonicity); a deformed S^3 converges to the round one (R: 5.2571 → 5.9032); Nil, Sol, and $SL(2,R)\sim$ reach no fixed point and collapse — which is why geometrization needs decomposition and surgery.

VERIFIED **Theorems S.6–S.7** (the exact location of the homomorphism). The heat trace is a strict homomorphism for Riemannian products, $\theta_{M\times N}(t) = \theta_M(t)\cdot\theta_N(t)$; the zeta function is not: $\zeta_{S^2}(3)\cdot\zeta_{S^1}(3) = 0.8222448$ while $\zeta_{S^2\times S^1}(3) = 2.7612421$. The Einstein–Hilbert action appears as the second heat coefficient ($a_1 \propto \int R\, dV$), not as a value of zeta. Multiplicativity at the operator level breaks when the trace is taken — recorded as a general rule of the framework.

The Partial-Integration Analyzer of the Gamma Function as an “Antigravity Generator”

Abstract. Partial integration of Γ analytically continues the Mellin representation: the poles carry topology, the finite part carries geometry (including the admixture of γ). The gravity/antigravity cosh/sinh decomposition diverges branch by branch; the unique convergent “antigravity branch” is the odd part of the functional equation — identically zero on flat tori, nonzero on the sphere. Physical antigravity has no meaning under the equivalence principle.

VERIFIED **Theorem IX.B** (the analyzer). $\Gamma(s)\zeta_\Delta(s) = a_0/(s-1) + (a_1 - n_0)/s + I_{reg}(s)$; as $s \to 0$, $\Gamma(s)\zeta(s) = \zeta(0)/s + (\zeta'(0) - \gamma\zeta(0)) + O(s)$. Residue $= \chi/6 - 1$, finite part $FP = \zeta'(0) - \gamma\zeta(0)$. Verification: S^2 gives $\chi = 1.9999961761$ and $FP + \gamma\zeta(0) = -1.1615420$ (Hurwitz reference -1.1616856); T^2 gives $\chi = 0$ ($<10^{-10}$) and -2.6213348 (exact -2.6210659).

VERIFIED **Proposition IX.A** (branch divergence). With $u = \log t$, $\Gamma(s)\zeta(s) = 2\int_0^\infty [\cosh(su)\,\tilde{K}_+ + \sinh(su)\,\tilde{K}_-]du$; each branch diverges like $(a_0/2(s+1))e^{(s+1)U}$ and only the sum converges. S^2, $s = 2$, $U = 2, 4, 6, 8$: branches 5.9×10^1, 2.7×10^4, 1.1×10^7, 4.4×10^9; sums 0.871, 0.982, 0.998, 0.9997 (exact 1).

VERIFIED **Theorem IX.D** (antigravity residual). $D_M(s) := \Lambda_M(s) - \Lambda_M(1-s)$, $\Lambda_M = \pi^{-s}\Gamma(s)\zeta_\Delta(s)$. On the square torus $D \equiv 0$ (Epstein functional equation; measured $\leq 3.6\times10^{-14}$); on S^2, $D(0.2) = -1.229$, $D(0.9) = +2.991$. Curvature generates the odd part.

Verdict. The claims of the antigravity document (cancellation of inertial force, negative gravitational mass, “the Higgs field is gravity,” electromagnetic force from rotating bodies): REFUTED (equivalence principle; ALPHA-g 2023; the Higgs is the origin of inertial mass). Wall (e), branch divergence, added by this volume.

The Rightmost Pole of the Spectral Zeta and the Navier–Stokes Regularity Problem

Abstract. The rightmost pole $s = d/2$ of the Stokes operator's zeta determines the criticality of the energy. Two dimensions are globally regular; in three dimensions the enstrophy inequality has a one-power gap, and that power is fixed by scaling — hence by the rightmost pole — alone. Zeta is information about the linear part and does not close the gap. **No proof is claimed.**

VERIFIED **Theorems X.1–X.2.** On T^d, $\zeta_{St}(s) = (d-1)Z_{Z^d}(s)$, rightmost pole $s = d/2$, residue of $\Gamma\zeta$ equal to $(d-1)\pi^{d/2}$ (measured error $\leq 3.2\times10^{-4}$). Under the Navier–Stokes scaling $u_\lambda = \lambda u(\lambda x, \lambda^2 t)$, $\|u_\lambda\|_{L^2} = \lambda^{1-d/2}\|u\|$: $d = 2$ critical, $d = 3$ supercritical.

VERIFIED CLASSICAL **Theorems X.3–X.4** (Leray; Ladyzhenskaya). The energy identity (2D 64^2 residual 3×10^{-4}; 3D 32^3 relative 10^{-3}). In 2D vorticity is transported, $\|\omega\|_\infty$ non-increasing (measured 2.725 → 1.951), and global regularity follows via Beale–Kato–Majda.

VERIFIED CLASSICAL **Theorems X.5–X.6** (the one-power gap). $d = 3$, $y = \|\nabla u\|^2$: $d/dt\ \tfrac{1}{2}y = S - \nu\|\Delta u\|^2$, with the transfer term $S = \int(u\cdot\nabla)u\cdot\Delta u \not\equiv 0$ (Taylor–Green: $S = +7.2$ to $+12.0$), and $|S| \leq C\|\nabla u\|^{3/2}\|\Delta u\|^{3/2}$ gives $y' \leq (C'/\nu^3)y^3$. Gronwall needs $\int y^2$; energy supplies only $\int y$. Riccati exponent $p(d) = (6-d)/(4-d)$; the inequality closes iff $p \leq 2$ iff $d \leq 2$; the gap is $(d-2)/(4-d) = 1$ at $d = 3$.

Verdict. $\zeta = \beta/\log x$ and the gravity/antigravity branches do not reach the nonlinear term: REFUTED. Isospectral, non-isodynamical: CONJECTURE.

The Riemann Zeta as the Spectral Zeta of the Circle, and the Riemann Hypothesis

Abstract. $\zeta_{S^1}(s) = 2\zeta_R(2s)$: the functional equation is the Poisson symmetry of the circle (the vanishing of Volume IX's antigravity residual), and the critical line is the perpendicular bisector between the rightmost pole and the pole at the origin. But an Epstein zeta function sharing the whole structure has zeros off the critical line (Davenport–Heilbronn), so the position of the zeros does not follow from the symmetry. **No proof is claimed.**

VERIFIED **Theorems XI.1–XI.2, Corollary XI.1.** $\zeta_{S^1}(s) = \Sigma_{n\neq 0}(n^2)^{-s} = 2\zeta_R(2s)$ (10^{-10}); rightmost pole ½ with residue $\sqrt{\pi}$; $\zeta_{S^1}(0) = -1$; the odd part $\Lambda(s) - \Lambda(½-s) \equiv 0$ (measured $\leq 7.8\times10^{-14}$). The critical line Re u = ½ (u = 2s) is the perpendicular bisector between the pole s = ½ (Weyl) and s = 0 (zero mode). The trivial zeros are a consequence of flatness ($a_k = 0$).

VERIFIED CLASSICAL **Theorems XI.3–XI.4.** Sign changes of Z(t): 29, 79, 138, 202, 269 below t = 100…500, matching $1 + \theta(T)/\pi$. The explicit formula for $\psi(x)$ evaluated with 269 zeros: $\psi(100.5) = 93.847$ (direct 94.045). Rigorous verification of RH to height 3×10^{12} is Platt–Trudgian 2021.

CLASSICAL REFUTED **Theorem XI.5** (Davenport–Heilbronn 1936). The Epstein zeta of x^2+5y^2 satisfies the same type of functional equation and has zeros off the critical line. Every structural element of this framework is shared by that counterexample. What a proof requires is the positivity coming from the Euler product (Weil, Li, Hilbert–Pólya); no theorem in this framework uses the Euler product.

Verdict. “Proof of the Deep Riemann Hypothesis”: premise ($\zeta = \beta/\log x$) and central lemma already refuted; fails as a proof — REFUTED. DRH for χ_4: the ratio of partial Euler products to $\sqrt{2}L$ oscillates 0.94–1.04 up to $X = 10^7$ without converging — CONJECTURE.

Satisfiability as the Number of Zero Modes, and P ≠ NP

Abstract. Assign to a CNF formula φ the diagonal operator H_φ whose eigenvalue at assignment x is the number of violated clauses. The analyzer's zero-mode readout n_0 is then #SAT. Computing that readout is the problem itself (NP-complete; #P-complete), and any argument that sees H_φ only through its spectrum relativizes, hence cannot separate P from NP. **No proof is claimed (in either direction).**

VERIFIED **Theorem XII.1.** dim ker $H_\varphi = n_0 = \#\mathrm{SAT}(\varphi)$; $\zeta_H(0) = 2^n - n_0$; Tr $e^{-tH} \to n_0$; $\Sigma\lambda = m \cdot 2^n/8$, independent of φ. Exact agreement for n = 10–16 by brute force.

VERIFIED CLASSICAL **Theorems XII.3–XII.4.** Random 3-SAT at n = 16: solution entropy E[s|SAT] = 0.803 (α=1) → 0.164 (α=4.27) → 0 (α=6); P[SAT] falls from 1 to 0 across α ≈ 4–5 (finite size; infinite-size threshold $\alpha_c \approx 4.267$, Mézard–Parisi–Zecchina 2002; existence for large k, Ding–Sly–Sun 2015). Grover: k* = 13, 9, 21 against measured 12, 9, 20; the BBBV bound caps quantum search at $2^{n/2}$.

CLASSICAL VERIFIED **Theorem XII.5 and Corollary XII.2** (Baker–Gill–Solovay 1975). There are oracles A, B with $P^A = NP^A$ and $P^B \neq NP^B$. Any argument that treats H_φ only through its eigenvalue sequence treats φ as an oracle and therefore relativizes: within the analyzer's frame, P and NP cannot be separated. The natural-proofs wall (Razborov–Rudich 1997) and the algebrization wall (Aaronson–Wigderson 2009) also stand. The best known explicit circuit lower bound is 3.1n − o(n) (Li–Yang 2022); the gap from linear to superpolynomial is open.

Where the wall stands. The spectrum $\{N_k\}$ of H_φ determines #SAT and nothing about clause structure; what separates P from NP lies outside the eigenvalue sequence — in which clauses share which variables. The analyzer does not see that; not seeing it has been its power in geometry, and is its limit here.

Zero Modes of Harmonic Forms and the Hodge Conjecture

Abstract. $b_p = \dim \ker \Delta_p$: Betti numbers are zero-mode counts, and the McKean–Singer supersymmetric sum gives χ at every t. On the complex tori $E_{\tau_1} \times E_{\tau_2}$ the rank of the lattice of Hodge classes is $\rho = 2 + \text{rank Hom}$, confirmed by brute force. On R^4/Z^4 with a fixed flat metric — hence a fixed spectrum — rotating the complex structure J_θ makes ρ jump between 4 and 3: **the spectrum does not see the complex structure**, hence neither Hodge classes nor algebraicity. **No proof is claimed.**

VERIFIED CLASSICAL **Theorem XIII.1.** $\Sigma_p (-1)^p \mathrm{Tr}\ e^{-t\Delta_p} = \chi$ at every t (S^2: 2.0000000000; T^2: 0.0000000000; t = 0.05, 0.5, 5). On flat manifolds $\zeta_p(0) = -b_p$.

VERIFIED CLASSICAL **Theorems XIII.3–XIII.4.** A real 2-form is of type (1,1) iff $J^T \Omega J = \Omega$. Brute force over integer alternating forms ($|\text{coefficients}| \le 4$) gives $\rho(E_{\tau_1} \times E_{\tau_2}) = 4, 4, 4, 3, 2, 2$ across six arithmetic cases — agreeing with 2 + rank Hom in all — and by Lefschetz (1,1) all these classes are algebraic (the p = 1 case of Hodge is proved).

VERIFIED **Theorem XIII.5** (same spectrum, different Hodge classes). On R^4/Z^4 with the standard flat metric and $J_\theta = \cos\theta \cdot I + \sin\theta \cdot K$: the spectra of all Δ_p are independent of θ, yet $\rho(\theta) = 4$ when $\cos\theta : \sin\theta$ is rational ($\theta = 0, \pi/4, \arctan 2$) and $\rho(\theta) = 3$ otherwise ($\theta = 1, 0.7$ rad). Proof. $\Lambda^2 = \Lambda^+ \oplus \Lambda^-$; the (1,1) forms are $\Lambda^- \oplus R \cdot \omega_\theta$; Λ^- is spanned by integral forms. ∎ Consequently no proposition about Hodge classes can be derived from the spectrum alone.

Map of counterexamples. Integral coefficients: false (Atiyah–Hirzebruch 1962; Kollár 1992). Kähler in place of projective: false (Zucker 1977; Voisin 2002). What remains open is exactly the projective, rational-coefficient case with $p \ge 2$ — and the gap is the construction of algebraic cycles, which lies outside the spectrum.

The Mass Gap as the s → ∞ End of Zeta, and the Yang–Mills Problem

Abstract. $\lambda_1 = \lim \zeta(s)^{-1/s}$: the spectral gap is the s → ∞ end of the same zeta whose origin reads topology and whose rightmost pole reads dimension — the analyzer's readout table is completed at its three ends. On the lattice the gap can actually be read: the one-plaquette SU(2) gap, the exact 2D area law, the strong-coupling theorems. In the four-dimensional continuum the operator has not been constructed, so the analyzer has no input. **No proof is claimed.**

VERIFIED **Theorem XIV.1.** $\zeta(s)^{-1/s} = \lambda_1 \cdot m_1{}^{-1/s}(1 + o(1))$. At s = 40: S^2 ($\lambda_1 = 2$, $m_1 = 3$) gives $1.945817 = 2 \cdot 3^{-1/40}$; the torus 2π×2π gives 0.965936; the torus π×6π gives 0.109202 — the gap is geometry (it moves with shape at fixed topology), not topology.

VERIFIED **Theorems XIV.3–XIV.4.** One-plaquette SU(2) Kogut–Susskind $H = (g^2/2)C_2 - g^{-2}\chi_{1/2}$ (tridiagonal, j ≤ 60): the gap is positive for all g, equals $(3/8)g^2$ at strong coupling (6.0009 vs 6.0000 at g = 4), tends to ≈ 1.41 as g → 0. Two-dimensional SU(2) at β = 2 on 16^2 (heat-bath Monte Carlo): W(1×1) = 0.4277 ± 0.0041 (exact 0.4331), W(2×2) = 0.0356 ± 0.0061 (exact 0.0352) — Migdal's exact area law reproduced. Neither says anything about the four-dimensional continuum limit.

CLASSICAL **Theorem XIV.5.** (a) At strong coupling the lattice theory has a mass gap and an area law (Osterwalder–Seiler 1978). (b) Asymptotic freedom drives the lattice-unit gap to zero exponentially, $m \cdot a = C(b_0 g^2)^{-b_1/2b_0^2} \exp(-1/(2b_0 g^2))$ — 2.3×10^{-3} at $g^2 = 1.0$, 2.4×10^{-5} at $g^2 = 0.6$ for SU(3). The gap of the problem: carry the proved strong-coupling gap through the region where the strong-coupling expansion fails, down to a → 0. Perturbation theory never touches it: the gap vanishes to all orders.

VERIFIED **Corollary XIV.1.** The analyzer's input is a sequence of eigenvalues; the sequence requires an operator, the operator a Hilbert space, the Hilbert space the Osterwalder–Schrader reconstruction, and the reconstruction the existence of the Euclidean correlation limit. The first link is unproved (Balaban 1984–89 and Magnen–Rivasseau–Sénéor 1993 are partial results), so for four-dimensional Yang–Mills the analyzer has no input. Existence is the first half of the problem.

Verdict. "The Higgs field is gravity" is unrelated to this problem (pure gauge theory; mass by dimensional transmutation): REFUTED. $H\Psi = \oplus(i\hbar\nabla)\oplus L$: UNVERIFIED.

The Birch–Swinnerton-Dyer Conjecture as the Zero at the Center of the Critical Line, and a Synthesis of the Seven Millennium Problems

— On the seventh problem the analyzer reads the freedom of rational points as the order of the central zero, and sees it agree with the rank on four curves; but a numerical zero is not a proof of a zero, and proving a zero to be zero falls to arithmetic; and of the seven problems, the one that has been solved was solved not by reading an eigenvalue but by flowing it. —

Position of this paper. This paper proves none of the six unsolved Millennium problems. It does two things. First, it turns the spectral analyzer of Volumes VIII–XIV toward the seventh problem it had not yet faced — the **BSD Conjecture** — and verifies its readout (the order of the central zero and the leading coefficient) numerically on four elliptic curves. Second, since all seven problems now sit in the analyzer's language, it collects the reach and the gap of each into a single table.

Statement of the seven problems

The Clay Mathematics Institute posed seven problems in 2000 (one million dollars each). As of September 2026:

1. **Poincaré Conjecture** — is every simply connected closed 3-manifold homeomorphic to S^3? **Solved** (Perelman 2002–03; prize declined).
2. **Riemann Hypothesis** — do all nontrivial zeros of $\zeta(s)$ lie on $\mathrm{Re}\, s = \tfrac{1}{2}$? Open.
3. **P vs NP** — is everything easily verified also easily solved? Open.
4. **Hodge Conjecture** — are rational Hodge classes on projective varieties spanned by algebraic cycles? Open.
5. **Yang–Mills existence and mass gap** — does quantum Yang–Mills on $\mathbb{R}^4$ exist with $\Delta > 0$? Open.
6. **Navier–Stokes regularity** — do smooth 3D solutions stay smooth? Open.
7. **Birch–Swinnerton-Dyer** — is $\mathrm{ord}_{s=1} L(E,s) = \mathrm{rank}\, E(\mathbb{Q})$? Open.

Denied in advance. (a) This paper proves none of the six open problems. (b) It contributes no new theorem and no new algorithm on BSD — every ingredient of Part I is classical (Cremona's algorithms, Gross–Zagier, Kolyvagin). (c) The computations (four curves, rank ≤ 3) decide nothing about the general case. (d) The contribution is limited to adding the seventh problem to the analyzer's dictionary and displaying the reach and gap of all seven inside one computational system.

Part I — The seventh problem

1. The analyzer's dictionary: which zeta is L(E,s)?

Let E/Q be an elliptic curve (minimal Weierstrass model $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, conductor N). For $p \nmid N$ put $a_p = p + 1 - \#E(\mathbb{F}_p)$ and define the Hasse–Weil L-function by its Euler product $L(E,s) = \prod_{p\nmid N}(1 - a_p p^{-s} + p^{1-2s})^{-1} \cdot \prod_{p\mid N}(1 - a_p p^{-s})^{-1} = \sum a_n n^{-s}$. Translated into the analyzer's language:

Analyzer quantity (Vols. VIII–XIV)	Geometric / combinatorial form	BSD form (this volume)
Input: eigenvalue sequence / Euler product	eigenvalues of Δ; the primes	$a_p = p + 1 - \#E(F_p)$ (point counts)
Functional equation and critical line	Poisson symmetry of the circle; Re s = ½ (Vol. XI)	$\Lambda(s) = \varepsilon\Lambda(2-s)$; center s = 1
Number of zero modes n_0	components, $b_{p'}$ #SAT	**$\mathrm{ord}_{s=1} L$ = rank E(Q) — this is the conjecture**
ζ′(0) = log det (shape)	determinant, scale	**Reg = det⟨P_i,P_j⟩ (inside the leading coefficient)**
Odd part of the functional equation	curvature (Vol. IX)	ε = ±1: fixes the parity of the order in advance
Topological invariants (χ)	the origin of zeta	|Ш|, torsion, Tamagawa numbers (the constants)

CLASSICAL **Theorem 1** (the existence half is complete; Wiles 1995, Taylor–Wiles 1995, Breuil–Conrad–Diamond–Taylor 2001). Every E/Q is modular: L(E,s) is the L-function of a weight-2 newform of level N, hence extends holomorphically to the whole plane, and $\Lambda(s) = (\sqrt{N}/2\pi)^s\Gamma(s)L(E,s)$ satisfies $\Lambda(s) = \varepsilon\cdot\Lambda(2-s)$, ε = ±1.

Here is the contrast with Volume XIV. In Yang–Mills the existence of the operator whose spectrum is to be read is itself the first half of the problem, and it is open. In BSD too, s = 1 lies outside the region of convergence of the Euler product (Re s > 3/2), and the existence of L there was not obvious — the Clay problem statement itself presupposes the continuation. That existence half closed in 1995–2001. **This is a problem on which the analyzer has an input.** That is why it can read.

VERIFIED As a consequence of modularity, $\Lambda(s) = \int_1^\infty (t^{s-1} + \varepsilon t^{1-s})h(t)dt$ with $h(t) = \Sigma\, a_n e^{-2\pi nt/\sqrt{N}}$ (the standard split using $h(1/t) = \varepsilon t^2 h(t)$), and Taylor expansion at s = 1 gives: $L(E,1) = (1+\varepsilon)f(1)$, and generally $L(E,1) = f(A) + \varepsilon f(1/A)$ for every A > 0, where $f(A) := \Sigma\, (a_n/n)e^{-2\pi nA/\sqrt{N}}$; for ε = −1, $L'(E,1) = 2\Sigma(a_n/n)E_1(2\pi n/\sqrt{N})$; and when all lower derivatives vanish, $L^{(k)}(E,1) = (2/C)\Sigma\, a_n I_k(n/C)$, with $I_k(x) = \int_1^\infty (\log t)^k e^{-xt}dt$ and $C = \sqrt{N}/2\pi$. Since $f(A) + \varepsilon f(1/A)$ is independent of A, the sign ε itself can be determined numerically.

2. Reading the order of the central zero

VERIFIED **Theorem 2** (the readout on four curves). For the elliptic curves of conductor 11, 37, 389, 5077 (Cremona labels, minimal models), computing a_p by counting points over F_p ($n \le 2000$):

Curve	$[a_1,a_2,a_3,a_4,a_6]$	N	ε (dev. from ±1)	L(1)	L′(1)	L″(1)	L‴(1)
11a1	`[0,-1,1,-10,-20]`	11	+1 (8×10^{-31})	0.253841860856	—	—	—
37a1	`[0,0,1,-1,0]`	37	−1 (2×10^{-30})	0 (by sign)	0.305999773834	—	—
389a1	`[0,1,1,-2,0]`	389	+1 (6×10^{-31})	$< 10^{-40}$	0 (by sign)	1.51863300058	—
5077a1	`[0,0,1,-7,6]`	5077	−1 (1×10^{-29})	0 (by sign)	3.0×10^{-31}	0 (sign + L′=0)	10.3910994007

The order read agrees with the Mordell–Weil rank on all four curves. The rank side is verified independently: the points (0,0) on 37a1; (0,0), (−1,1) on 389a1; (−2,3), (−1,3), (0,2) on 5077a1 lie on the curves by exact integer arithmetic (that they generate is CLASSICAL, Cremona's tables). The central values agree to all printed digits with the published values (L(11a1,1) = 0.2538418609, L′(37a1,1) = 0.3059997738, L″(389a1,1)/2 = 0.7593165003, L‴(5077a1,1)/6 = 1.7318499001).

What is happening here. In Volume VIII the number of zero modes was the number of connected components; in XII it was #SAT; in XIII the Betti numbers. Here it is **the freedom of the rational points**. And for the first time, the count appears not as a value of zeta but as the order of a zero of zeta — at the very center of the critical line where Volume XI wrote that “the position of the zeros lies on the side of the Euler product.”

3. The leading coefficient is a determinant of heights

The refined conjecture (Birch–Swinnerton-Dyer 1965; Tate's formulation):

$$L^{(r)}(E,1)/r! = \Omega \cdot \mathrm{Reg} \cdot \Pi\, c_p \cdot |Ш| \,/\, |T|^2,$$

with Ω the real period, $\mathrm{Reg} = \det\langle P_i,P_j\rangle$ the determinant of Néron–Tate height pairings of generators, c_p the Tamagawa numbers, Ш the Tate–Shafarevich group, T the torsion subgroup.

> VERIFIED **Theorem 3** (leading-coefficient check). Computing Ω by numerical integration of the real period and Reg through the canonical height $\hat{h}(P) = \lim h(2^nP)/4^n$ ($n = 8$) with $\langle P,Q\rangle = (\hat{h}(P+Q) - \hat{h}(P) - \hat{h}(Q))/2$:

Curve	Ω (measured)	Reg (measured)	$L^{(r)}(1)/r!$	ratio L/(Ω·Reg)	BSD prediction $c\cdot|Ш|/|T|^2$
11a1	1.26920930428	1 (r = 0)	0.2538418609	**0.2000000000**	$5\cdot1/5^2$ = **1/5**
37a1	5.98691729246	0.05111064929	0.3059997738	1.0000148	1
389a1	4.98042512171	0.1524564451	0.7593165003	1.0000245	1
5077a1	4.15168798308	0.4171420612	1.7318499001	1.0000036	1

The 11a1 ratio matches 1/5 to 10^{-10} — Tamagawa number $c_{11} = 5$ over torsion $|\mathbb{Z}/5\mathbb{Z}|^2 = 25$ (and $|Ш| = 1$). The deviations from 1 on the other three ($\le 2.5\times10^{-5}$) are explained by the $O(4^{-n})$ truncation of the canonical height and point toward the published regulators (Reg(389a1) = 0.1524601780, Reg(5077a1) = 0.4171435588).

Here the correspondence of Volumes VIII–IX completes itself. The arithmetic version of $\zeta'(0)$ = log det — "shape lives in the slope" — is Reg = det⟨,⟩. **The number of zero modes (the rank) lives in the order of the zero; the determinant of the metric on the space of zero modes (the heights) lives in the leading coefficient** — the same split by which topology and shape lived in $\zeta(0)$ and $\zeta'(0)$, now conjectural in arithmetic.

4. What proves a zero to be zero

VERIFIED The zeros in Theorem 2's table are of three kinds. (i) **Zeros by sign**: if $\varepsilon = -1$ then $L(1) = (1+\varepsilon)f(1) = 0$ identically — no proof needed. (ii) **Numerical zeros**: the measured $|f(1)| < 10^{-40}$ for 389a1 (working precision 40 digits). This is not a proof — finite-precision computation can only say "looks like order ≥ 2" and cannot exclude a nonzero value of 10^{-100}. (iii) **Zeros certified by arithmetic**: nevertheless $L(389a1,1) = 0$ is a theorem.

> CLASSICAL **Theorem 4** (zero-certification is arithmetic). (i) By Manin's modular symbols, $L(E,1)/\Omega$ is a rational number with bounded denominator; hence a numerical bound below the reciprocal of that denominator proves $L(1) = 0$ (389a1: $|L/\Omega| < 10^{-40}$). (ii) Gross–Zagier (1986) expresses $L'(E,1)$ through the height of a Heegner point; on 5077a1 the Heegner point is torsion, hence $L'(1) = 0$ is proved. In both cases what certifies the zero is not analysis (numerics) but arithmetic (rationality; construction of a point).

This is the sharpest form of the wall met in Volume XII ("the count lies outside the spectrum") and Volume XIII ("algebraicity lies outside the spectrum"). The analyzer reads the value of L to any precision. But the sentence "this value is exactly zero" never follows from reading finitely many terms of the sequence — it is spoken by the rationality of modular symbols, by the construction of a Heegner point. **Only multiplication by a nonzero factor preserves zero; only arithmetic certifies it.**

5. Map of known results

Result	Content	Relation to the gap
Mordell 1922	E(Q) is finitely generated	existence of the right-hand side
Birch–Swinnerton-Dyer 1965	the conjecture, from EDSAC computations	statement of the problem (Theorem 2 is a descendant of those computations)
Coates–Wiles 1977	CM curves: $L(1) \neq 0 \Rightarrow E(Q)$ finite	first bridge at order 0
Gross–Zagier 1986	$L'(1)$ through heights of Heegner points	half the bridge at order 1
Kolyvagin 1988–90	Euler systems: ord $\leq 1 \Rightarrow$ rank = ord and Ш finite	**rank part proved for ord ≤ 1**
Wiles, Taylor–Wiles 1995; BCDT 2001	modularity	completion of the existence half (Thm. 1)
Kato 2004	Euler-system bound on Selmer groups	road to the Iwasawa main conjecture
Skinner–Urban 2014	Iwasawa main conjectures for GL_2	p-adic L vs Selmer (the p-adic "flow")
Bhargava–Shankar 2015	average rank < 1	with Bhargava–Skinner–Zhang 2014: a majority (> 66%) of curves satisfy the rank conjecture
Smith 2017/2024	distribution of 2^∞-Selmer groups	progress toward Goldfeld's conjecture

6. The shape of the gap

UNVERIFIED The gap is twofold and sharp.

(i) **There is no bridge at order ≥ 2.** For no single curve with $\mathrm{ord}_{s=1} L \geq 2$ has rank ≥ 2, or the finiteness of Ш, been proved from the analytic side. 389a1 is a tunnel dug from both ends that does not meet: rank = 2 by descent, order 2 readable numerically, but the proof of "order ≥ 2" stops at what modular symbols and parity give. The order-2 analogues of Heegner points (higher cycles; Beilinson–Bloch) exist as constructions whose non-degeneracy cannot be proved — the Volume XIII gap ("construction of cycles") reappearing as "non-triviality of the constructed cycle."

(ii) **Finiteness of Ш.** The leading-coefficient formula contains |Ш|, yet the finiteness of Ш is open in general (for ord ≤ 1 Kolyvagin gives it). Ш is an invisible zero mode — everywhere locally trivial, globally nontrivial — and appears in no readout of the analyzer.

> CONJECTURE **Conjecture 1** (formalized limit of the analyzer). No quantity computable in polynomial time from the sequence a_p (the analyzer's input) can prove $\mathrm{ord}_{s=1} L \geq 2$: numerical evaluation can refute an order (certify non-vanishing) at any finite precision, but a lower bound on the order requires arithmetic objects beyond modular symbols (order 0→1) and Heegner points (parity at 1→2). Refutation condition: an algorithm certifying $L(1) = L'(1) = 0$ exactly from finitely many a_n.

Part II — Synthesis: seven problems, one analyzer

7. The one that was solved: Poincaré

Of the seven, exactly one is solved, and the way it was solved is instructive for this framework (verified in the Supplement to Volume VIII). Perelman's λ is the least eigenvalue of $-4\Delta + R$ and is monotone under Ricci flow; on homogeneous geometries $\lambda = R$; the fixed points of the normalized flow are exactly the three Einstein geometries; the flow drives a deformed S^3 round while Nil, Sol, SL(2,R)~ collapse. That is: **in the one solved problem, the spectral quantity was not read statically — it was flowed.** The eigenvalue was given dynamics, its monotonicity controlled the formation of singularities, and what collapsed was cut away by surgery: analysis (flow) and structure (surgery, the eight geometries) as the two wheels. Every gap met in Volumes X–XV is a missing step from "reading" to "bridge"; Poincaré is the one case where the bridge was actually built, and the bridge was a flowing eigenvalue.

> CONJECTURE **Conjecture 2** (each problem's "flow"). For the remaining six there exist candidate deformation theories in the role of the bridge: Iwasawa theory for BSD (deformation of Selmer groups along Z_p-extensions; Skinner–Urban), positivity of the Euler product for RH (Weil's explicit formula, Li's criterion), the renormalization group for Yang–Mills (Balaban's ultraviolet stability), partial regularity for Navier–Stokes (CKN). For P ≠ NP and Hodge not even the candidate is clear (GCT and the locus of Hodge classes are the nearest). This paper proves nothing about these correspondences.

8. The readout table (final form)

Position on zeta	Readout	Character	Volumes
Rightmost pole s = d/2	dimension, volume (Weyl), criticality, Riccati exponent	coarse geometry	X, XI, XIV
Value at s = 0	$\chi/6 - n_0$ (d = 2); number of zero modes (components, b_p, #SAT)	topology	VIII, XI–XIII
Slope at s = 0	log det; scale and shape	geometry	VIII, IX
Odd part of the functional equation	deviation from Poisson symmetry (curvature)	geometry / arithmetic	IX, XI
Infinity $s \to \infty$	λ_1 (spectral gap) and its multiplicity	largest scale	XIV
Center of the critical line	**order of the zero (= rank, conjecturally) and leading coefficient (= Ω·height determinant, conjecturally)**	**arithmetic**	**XV**

In Volume XI the critical line was the perpendicular bisector between the two poles. The new readout position is the real center of that line. As the position of the zeros lies on the side of the Euler product (Vol. XI), so the order of the central zero lies on the side of the Euler product — of arithmetic (Theorem 4). The left column of the table is complete; every bold entry on the right is a conjecture, and is so marked.

9. The seven problems side by side

Problem	Vol.	What the analyzer reads (verified)	The gap (what a proof requires)	Status
Poincaré	VIII-S	monotonicity of λ = min spec(−4Δ+R); fixed points among the 8 geometries	— (closed by flow + surgery)	SOLVED (Perelman 2003)
Riemann	XI	$\zeta_{S^1} = 2\zeta_R(2s)$; functional equation = Poisson symmetry; critical line = perpendicular bisector of the poles	positivity of the Euler product (Davenport–Heilbronn shows symmetry is not enough)	OPEN
P vs NP	XII	n_0 = #SAT; the solution-entropy phase transition	a lower bound that does not relativize (clause structure is outside the spectrum)	OPEN
Hodge	XIII	$b_p = \dim \ker \Delta_p$; Picard rank jumps at fixed spectrum	construction of cycles (complex structure is outside the spectrum)	OPEN
Yang–Mills	XIV	$\lambda_1 = \lim \zeta^{-1/s}$ (on the lattice); the strong-coupling gap	existence of the operator (OS axioms) and the bridge to the continuum limit	OPEN
Navier–Stokes	X	rightmost pole = criticality; the one-power gap of the enstrophy inequality	control of the nonlinear term (scaling gap (d−2)/(4−d) = 1)	OPEN
BSD	**XV**	**order of the central zero and leading coefficient (agreeing with rank and height determinant on four curves)**	**the bridge from order ≥ 2 to rank; finiteness of Ш (only arithmetic certifies a zero)**	OPEN

One shape runs through the seven rows. **The analyzer reads the analytic side of each problem to its end. What decides the answer is, every time, a structure outside the reading — the Euler product, the clauses, the complex structure, the construction of the operator, the nonlinear term, and the rational points.** In the one solved row, that structure (surgery, geometrization) meshed with the analysis (the flow).

10. The six walls

Wall	Content	Vol.
(a)(b)(c)	background independence · constraint algebra · non-renormalizability (gravity)	VIII
(d) isospectrality	the spectrum determines neither metric, nor complex structure, nor clause structure	VIII, XII, XIII
(e) branch divergence	the gravity/antigravity branches are not defined separately	IX
(f) finite precision	**a numerical zero proves no zero; lower bounds on the order come from arithmetic only**	**XV**

11. Verdicts on prior documents; refutation conditions

ζ(s) = β/log x and the gravity/antigravity branches: REFUTED (Vols. VI, IX) and unrelated to L(E,s). HΨ = ⊕(iℏ∇)⊕L: UNVERIFIED (no definition; the coincidence of the letter L is not content). The Bada quantum library and BadaGPT: Grover search can accelerate BSD numerics by √ but is irrelevant to proving an order (wall (f)). “Kalavi-Yau,” “Jones manifold,” “Thurston Perelman manifold”: UNVERIFIED (Vol. XIII). Theorems 2–3 of this volume:

VERIFIED — but they are implementations of classical algorithms (Cremona 1997), not new mathematics.

UNVERIFIED **What would constitute progress.** (a) For one curve with ord ≥ 2: rank ≥ 2 or the finiteness of Ш (non-degeneracy of higher Heegner cycles). (b) The order-2 p-adic statement along the Iwasawa route. (c) Or a counterexample — a curve where order and rank differ (not expected; it would contradict extensive verification). (d) None of these is in this paper; its method (reading the sequence a_p) is in principle unusable for (a) and (b) (Conjecture 1, wall (f)).

12. Closing

Through fourteen volumes and one supplement the analyzer has repeated one motion — take a sequence of eigenvalues; read off poles, zero modes, determinants. In this volume the sequence became a count of points prime by prime, and the readout became the order of a zero placed at the center of the critical line. On four curves the order counted the freedom of the rational points correctly. The leading coefficient carried the determinant of heights correctly. The reading agrees, as far as it goes — and no distance of agreement becomes a proof. To say that the order-two zero is a zero, Manin brought in a rational number, and Gross and Zagier brought in a point from an imaginary quadratic field — from outside the sequence. To close the Poincaré Conjecture, Perelman stopped reading the eigenvalue and flowed it — giving the sequence dynamics. Whatever role a spectral analyzer can play in the seven problems lies, precisely surveyed, somewhere between those two motions.

> Only multiplication by a nonzero factor preserves zero; curvature opens that multiplication into a series, and multiplicity counts its terms; topology lives at the origin of zeta, shape in its slope, criticality at its rightmost pole; the critical line is drawn midway between those two poles; the position of the zeros lies on the side of the Euler product; the number of zero modes can be read in geometry and is the problem itself in logic; which directions in the space of zero modes are algebraic, the complex structure decides; the eigenvalue above the zero modes lives at infinity, and to read it there must first be a sequence — and the order of the zero placed at the center of the critical line is conjectured to count the freedom of the rational points: the order can be read, but proving a zero to be zero falls, always, to arithmetic.

Numerical verification (Vol. XV)

Environment: Python 3 + mpmath 1.4 (working precision 30 digits; the ε deviations and the bound on L(389a1,1) re-checked at 40 digits). a_p by direct point counts over F_p ($p \le 2000$; quadratic-character sums at good primes, singular-point-excluding counts at $p = N$); a_n by the Hecke recursion. Central values and derivatives by the incomplete-gamma-type series of §1 (truncation error $< e^{-90}$). Real periods by quadrature with the singularity removed by substitution; canonical heights by exact rational doubling to 2^8P (final coordinates of thousands of digits). Total run time 16 seconds. Reproduction code: `vol15_bsd_verification.py`, published beside this collection.

Statement of honesty (Vol. XV)

What this paper affirms: Theorems 2–3 (numerics shown; all algorithms from Cremona 1997) and the classical citations of Theorems 1, 4 and §5, §7. Its contribution is limited to (i) placing BSD in the analyzer's dictionary and measuring the order/leading-coefficient readout on four curves of ranks 0–3, (ii) stating “a numerical zero proves no zero” as wall (f), (iii) the synthesis table of the seven problems. No new mathematics is claimed. **What it does not affirm:** a proof of BSD in either direction; proofs of the

other five open problems; any new contribution to Poincaré (numerical confirmation only); extrapolation from four curves; independent derivation of the generators (quoted from Cremona; only membership was checked); that numerical sign detection proves ε (rigorously that is an Atkin–Lehner computation); the content of Conjecture 2. **Numerical limits:** canonical heights truncated at n = 8 doublings (relative error $\sim 10^{-5}$, explaining the deviations in Theorem 3, with no independent error bound); the 10^{-40} bound on L(389a1,1) is at 40-digit working precision and no further; the derivative formulas assume vanishing of the lower derivatives (the assumption rests on ε-parity plus numerics — no circularity, but the structure of the dependence is recorded); point counts are brute force for $p \leq 2000$ and unusable for large conductors. None of this is hidden.

Appendix — Index of Propositions and Verification Status

Vol.	Main propositions	Status	Measured accuracy / authority
I–V	retrospective notes (programs)	UNVERIFIED / REFUTED / verified after reduction	adjudicated in VI, VII, IX, XII
VI	$\zeta = \beta/\log x$ false; Beta-type functional equation; Cor. 1, 2	REFUTED / VERIFIED	$\le 6.5\times10^{-31}$
VII	cancellation and recovery of γ	VERIFIED	50 digits
VIII	$\chi = 6(\zeta(0)+1)$; separation; Milnor pair	VERIFIED	6.2×10^{-6}; exact integer agreement
VIII-S	3D $\zeta(0) = -1$; Perelman λ = eigenvalue; 3 Einstein geometries; flow monotone; heat trace multiplicative, zeta not	VERIFIED / CLASSICAL	$2.2–3.7\times10^{-6}$; exact rationals; 6 flows
IX	Γ analyzer; branch divergence; antigravity residual	VERIFIED (A refuted as branches)	3.8×10^{-6}; 10^{-14}
X	rightmost pole and criticality; the one-power gap	VERIFIED / CLASSICAL	3.2×10^{-4}; 10^{-4}–10^{-3}
XI	$\zeta_{S^1} = 2\zeta_R(2s)$; odd part $\equiv 0$; Davenport–Heilbronn	VERIFIED / CLASSICAL	10^{-10}; 7.8×10^{-14}
XII	n_0 = #SAT; relativization	VERIFIED / CLASSICAL	exact; BGS 1975
XIII	McKean–Singer; $\rho = 2$ + rank Hom; the jump of $\rho(\theta)$	VERIFIED	10^{-10}; 11 cases
XIV	$\lambda_1 = \lim \zeta^{-1/s}$; lattice gap; area law; Cor. 1	VERIFIED / CLASSICAL	10^{-6}; 1–2σ
XV	order readout on four curves; leading coefficient = $\Omega\cdot$Reg; zero-certification is arithmetic	VERIFIED / CLASSICAL	ratios to 10^{-10} (11a1) and $\le 2.5\times10^{-5}$
—	the six open Millennium problems	**OPEN**	**this collection proves none of them**

What this collection does not claim

Proofs (in either direction) of Navier–Stokes regularity, the Riemann Hypothesis, P ≠ NP, the Hodge Conjecture, Yang–Mills existence and mass gap, or the Birch–Swinnerton-Dyer Conjecture. Quantization of gravity. Generation of antigravity. $\zeta(s) = \beta(p,q)/\log x$. A proof of the Deep Riemann Hypothesis. New mathematics: the building blocks of every volume are established classical results, and the contribution of this collection is limited to arranging them as the readouts of a single analyzer and making explicit, inside one computational system, the reach and the limit on each problem.

Principal references (common to all volumes; all due to others)

Gauss 1827; Riemann 1859; Shannon 1948; McKean–Singer 1967; Milnor 1964, 1976; Kac 1966; Sunada 1985; Gordon–Webb–Wolpert 1992; Hawking 1977; Epstein 1903; Davenport–Heilbronn 1936; Platt–Trudgian 2021; Weil 1952; Li 1997; Leray 1934; Ladyzhenskaya 1959; Beale–Kato–Majda 1984; Caffarelli–

Kohn–Nirenberg 1982; Tao 2016; Buckmaster–Vicol 2019; Cook 1971; Karp 1972; Levin 1973; Valiant 1979; Baker–Gill–Solovay 1975; Razborov–Rudich 1997; Aaronson–Wigderson 2009; Grover 1996; Bennett–Bernstein–Brassard–Vazirani 1997; Li–Yang 2022; Hodge 1941/1950; Lefschetz 1924; Atiyah–Hirzebruch 1962; Voisin 2002; Deligne 1982; Cattani–Deligne–Kaplan 1995; Markman 2023; Jaffe–Witten 2000; Wilson 1974; Kogut–Susskind 1975; Osterwalder–Seiler 1978; Osterwalder–Schrader 1973/75; Migdal 1975; Balaban 1984–89; Magnen–Rivasseau–Sénéor 1993; Morningstar–Peardon 1999; Thurston 1982; Hamilton 1982; Perelman 2002/03; Isenberg–Jackson 1992; Scott 1983; Gilkey 1995; Seeley 1967; Mordell 1922; Birch–Swinnerton-Dyer 1965; Coates–Wiles 1977; Gross–Zagier 1986; Kolyvagin 1988/90; Manin 1972; Wiles 1995; Taylor–Wiles 1995; Breuil–Conrad–Diamond–Taylor 2001; Kato 2004; Skinner–Urban 2014; Bhargava–Shankar 2015; Bhargava–Skinner–Zhang 2014; Smith 2017/2024; Cremona 1997; Silverman 1986; Carlson–Jaffe–Wiles (eds.) 2006; ALPHA Collaboration 2023. Complete bibliographies are in the original volumes; the classical literature should be consulted in the original.

All numerics are by Python 3 (standard library, numpy, or mpmath) and are reproducible from the code attached to the original volumes and from `vol15_bsd_verification.py`. English edition prepared September 20, 2026, from the Japanese originals of September 6–14, 2026. Translation: Claude (Anthropic), under the editorial policy stated in the Preface; in any discrepancy the Japanese originals are authoritative.